Mirror Descent with Implicit Bias in Deep Learning

DS 608 Paper Presentation

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Introduction

The Rise of Deep Learning

The 2012 Moment

- AlexNet wins ImageNet 2012 by a 10% margin over second place — the moment the field took notice

What Made It Work?

- Massive overparameterization ($p \gg n$), SGD training, and empirical regularization techniques (dropout, batch norm)

But this raised a puzzle: classical learning theory predicted catastrophic overfitting.

The Generalization Mystery: Zhang et al. (2017)

Same architecture, same optimizer, same hyperparameters. Only the labels changed:

Model	Condition	Train Acc.	Test Acc.
Inception	True labels	100%	∼87%
Inception	Random labels	100%	∼10%
Inception	Random pixels	100%	∼10%
AlexNet	True labels	100%	∼80%
AlexNet	Random labels	100%	∼10%
MLP	True labels	100%	∼53%
MLP	Random labels	100%	∼10%

CIFAR-10. Zhang et al., ICLR 2017.

Model, size, optimizer, regularizer are identical across rows. Only labels differ. None of these explain generalization.

Classical Theory Cannot Explain This

Classical Generalization Bound

$$\text{Test error} \leq \text{Train error} + \underbrace{f(\text{model complexity}, n)}_{\text{complexity penalty}}$$

- Since networks fit *random* labels, their effective VC dimension / Rademacher complexity is essentially unbounded
- Therefore the complexity penalty is vacuously large — the bound predicts the network *should not* generalize even on true labels

The Implication

Networks memorize noise yet generalize on real data. This contradiction vanishes if the optimizer itself creates the bias.

Implicit Bias: The Optimizer as Regularizer

Setup: In the overparameterized regime, the solution set $\mathcal{W}$ is a high-dimensional manifold — infinitely many zero-loss solutions exist.

Known Result (Gunasekar et al., 2018)

For linear models, SGD converges to minimum ℓ_2 norm interpolant by geometry, not explicit constraint.

Three open questions that this paper answers:

1. What is the implicit bias for optimizers *beyond* SGD?
2. Does this extend to *nonlinear* models like deep networks?
3. Can we *deliberately engineer* the implicit bias by choosing the optimizer?

If (3) is yes, the optimizer is a design choice for generalization.

A Unified View: Stochastic Mirror Descent

SGD is just one special case

SMD generalizes all gradient-based optimizers via a potential function ψ

Potential $\psi(w)$	**Algorithm**	**Implicit Bias**
$\frac{1}{2}\|w\|_2^2$	SGD	Min ℓ_2 norm
$\sum_j w_j \log w_j$	Exponentiated Gradient	Sparsity / simplex
$\frac{1}{p}\|w\|_p^p$	p-norm algorithm	Min ℓ_p norm

The choice of ψ is the choice of bias. This paper proves which solution SMD finds for any ψ in nonlinear models.

Problem Setup and Notation

Overparameterized Nonlinear Models

- **Dataset:** $\{(x_i, y_i)\}_{i=1}^n$ where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$.
- **Model:** $f(x_i, w) = f_i(w)$ with parameters $w \in \mathbb{R}^p$.
- **The Interpolating Regime:** $p \gg n$.

The Set of Global Minima (Interpolants)

The set of all parameters consistent with the training data:

$$\mathcal{W} = \{w \in \mathbb{R}^p \mid f_i(w) = y_i, \quad i = 1, \ldots, n\} \tag{1}$$

This forms a high-dimensional $(p - n)$-dimensional manifold.

The Learning Objective

Minimize empirical risk $L(w) = \sum_{i=1}^n L_i(w)$, where $L_i(w) = \ell(y_i, f_i(w))$. Every $w \in \mathcal{W}$ yields $L(w) = 0$.

Optimization via SGD

Standard training typically utilizes **Stochastic Gradient Descent (SGD)**:

$$w_i = w_{i-1} - \eta \nabla L_i(w_{i-1}), \quad i \geq 1 \tag{2}$$

- $L_i(\cdot)$ is differentiable and non-negative.
- $L_i(w) = 0$ iff $y_i = f_i(w)$.
- For $i > n$, we cycle through the data or select at random.

The Implicit Question

Among all $w \in \mathcal{W}$, SGD will converge to exactly one. The update rule's geometry — not any explicit constraint — determines which one. This is the implicit bias.

Stochastic Mirror Descent (SMD)

The Geometry of Mirror Descent

Instead of updating in the weight space, SMD updates in the **mirrored domain** via a strictly convex potential function $\psi(\cdot)$.

The Update Rule

$$\nabla\psi(w_i) = \nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1}) \tag{3}$$

- Updates happen in dual space via $\nabla\psi$
- SGD: $\psi(w) = \frac{1}{2}\|w\|^2$ gives $\nabla\psi = \text{identity}$
- Different ψ encode different geometric biases

Every optimizer is mirror descent in disguise.

The Bregman Perspective

The SMD update can be equivalently viewed as a proximal minimization problem:

$$w_i = \arg\min_{w} \left\{ \eta w^T \nabla L_i(w_{i-1}) + D_\psi(w, w_{i-1}) \right\} \tag{4}$$

Bregman Divergence $D_\psi(w, \bar{w})$

Defined as the difference between the function and its first-order Taylor expansion:

$$D_\psi(w, \bar{w}) := \psi(w) - \psi(\bar{w}) - \nabla\psi(\bar{w})^T (w - \bar{w}) \tag{5}$$

- Non-negative: $D_\psi(w, \bar{w}) \geq 0$, equals 0 iff $w = \bar{w}$
- Tailored to geometry of ψ

Implicit Regularization in Linear Models

Consider the overparameterized linear model $f(x_i, w) = x_i^T w$.

Proposition 1: Convergence to the Closest Interpolant

For a sufficiently small step size η, the SMD iterates converge to:

$$w_\infty = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0) \tag{6}$$

- Out of infinite $\mathcal{W}$ solutions, SMD picks the Bregman-closest to initialization
- If $w_0 = \arg\min_w \psi(w)$, then $w_\infty = \arg\min_{w \in \mathcal{W}} \psi(w)$
- Geometry is the regularizer

Implication

First rigorous answer to Zhang et al.: optimizer encodes preference. This is determined by ψ and w_0.

Assumptions for Nonlinear Models

Assumption 1: Local Star-Convexity

Setup: Define region $\mathcal{B} = \{w \in \mathbb{R}^p \mid D_\psi(w, \bar{w}) \leq \epsilon\}$ containing w_0.

Assumption Statement: There exists $\bar{w} \in \mathcal{W}$ such that

$$D_{L_i}(\bar{w}, w) := L_i(\bar{w}) - L_i(w) - \nabla L_i(w)^T(\bar{w} - w) \geq 0$$

for all $w \in \mathcal{B}$ and $i = 1, \ldots, n$.

Geometric interpretation: The loss Bregman divergence is non-negative. This means the region near interpolating solutions exhibits local "star-convexity": curvature points inward toward the solution set.

Why it's reasonable: In the overparameterized regime ($p \gg n$), near any solution $\bar{w} \in \mathcal{W}$, the effective loss landscape becomes benign due to the abundance of degrees of freedom.

Assumption 2: Boundedness

Model $f_i(w)$ has bounded derivatives in $\mathcal{B}$.

- Gradient: $\|\nabla f_i(w)\| \leq \gamma$
- Hessian: $\alpha I \preceq H_{f_i}(w) \preceq \beta I$ for constants α, β

These prevent divergence under fixed step size η.

Main Theorems

Theorem 3: Deterministic Convergence

Under Assumption 1, and for a step size η such that $\psi(\cdot) - \eta L_i(\cdot)$ is strictly convex for all i, the following hold for the SMD iterates $\{w_i\}$:

1. **Containment:** All iterates $\{w_i\}$ remain within the region $\mathcal{B}$.
2. **Global Convergence:** The sequence $\{w_i\}$ converges to a limit w_∞.
3. **Interpolation:** The limit point is a global minimum, i.e., $w_\infty \in \mathcal{W}$.

Note: Deterministic convergence, no high-probability bounds needed.

Key point: Bregman geometry guarantees convergence independent of model.

Theorem 4: Characterization of the Limit Point

Let $w^* = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0)$ be the Bregman-closest interpolant to the initialization.

Proximity to the Optimal Regularized Solution

Under Assumptions 1 and 2, the limit point w_∞ satisfies:

1. $D_\psi(w_\infty, w_0) = D_\psi(w^*, w_0) + o(\epsilon)$
2. $D_\psi(w^*, w_\infty) = o(\epsilon)$

If w_0 is $O(\epsilon)$ from $\mathcal{W}$, then w_∞ is $o(\epsilon)$ from w^*.
Result: SMD implicitly minimizes ψ over interpolants.

Proof Technique: Fundamental Identity

Lemma 6: A General Identity for SMD

The authors derive a fundamental identity that holds for any model $f(\cdot,\cdot)$ and differentiable loss $\ell(\cdot)$. For any $w \in \mathcal{W}$:

$$D_\psi(w, w_{i-1}) = D_\psi(w, w_i) + D_{\psi-\eta L_i}(w_i, w_{i-1}) + \eta L_i(w_i) + \eta D_{L_i}(w, w_{i-1}) \tag{7}$$

Algebraic structure: All three RHS terms are non-negative (by convexity, non-negativity of loss, and star-convexity). This creates a telescoping inequality.

Takeaway: The Bregman distance to any interpolant is non-increasing, with three dissipative mechanisms:

1. Mirror-adjusted divergence shrapes solutions together
2. Loss terms vanish as iterates converge
3. Bregman loss divergence measures local geometry

Lemma 6: Proof Derivation

Start with Bregman definition: $D_\psi(w, w_i) = \psi(w) - \psi(w_i) - \nabla\psi(w_i)^T(w - w_i)$.
Substitute SMD update: $\nabla\psi(w_i) = \nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1})$:

$$D_\psi(w, w_i) = \psi(w) - \psi(w_i) - [\nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1})]^T(w - w_i)$$

Decompose by adding and subtracting $\psi(w_{i-1}) + \nabla\psi(w_{i-1})^T(w - w_{i-1})$:

$$D_\psi(w, w_i) = [D_\psi(w, w_{i-1}) - D_\psi(w_i, w_{i-1})] + \eta\nabla L_i(w_{i-1})^T(w - w_i)$$

Expand $(w - w_i) = (w - w_{i-1}) - (w_i - w_{i-1})$ and apply loss Bregman definitions.
Result: The fundamental identity with telescoping terms.

Lemma 6: Full Mathematical Derivation (Detailed)

Step 1. Expand $D_\psi(w, w_i)$ and $D_\psi(w_i, w_{i-1})$ separately, then combine:

$$D_\psi(w, w_i) = D_\psi(w, w_{i-1}) - D_\psi(w_i, w_{i-1}) + \eta \nabla L_i(w_{i-1})^T (w - w_i) \tag{8}$$

Step 2. For $w \in \mathcal{W}$, use $L_i(w) = 0$:

$$\eta \nabla L_i(w_{i-1})^T (w - w_{i-1}) = \eta L_i(w) - \eta L_i(w_{i-1}) - \eta D_{L_i}(w, w_{i-1})$$

(By definition of Bregman divergence for the loss.)

Step 3. Rearrange using $(w - w_i) = (w - w_{i-1}) - (w_i - w_{i-1})$:

$$\eta \nabla L_i(w_{i-1})^T (w - w_i) = -[L_i(w_{i-1}) - L_i(w_i) - \nabla L_i(w_{i-1})^T (w_i - w_{i-1})] - \eta D_{L_i}(w, w_{i-1})$$

$$= -D_{\psi - \eta L_i}(w_i, w_{i-1}) - \eta L_i(w_i) - \eta D_{L_i}(w, w_{i-1})$$

After rearrangement: the identity follows.

Proof of Theorem 3: Containment and Convergence

Part 1. Containment in $\mathcal{B}$: From Lemma 6 (Fundamental Identity):

$$D_\psi(w, w_{i-1}) = D_\psi(w, w_i) + D_{\psi-\eta L_i}(w_i, w_{i-1}) + \eta L_i(w_i) + \eta D_{L_i}(w, w_{i-1})$$

For $w \in \mathcal{W}$: all RHS terms are non-negative (by strict convexity of $\psi - \eta L_i$, sign of loss, star-convexity assumption). Thus $D_\psi(w, w_i) \leq D_\psi(w, w_{i-1}) \leq \cdots \leq D_\psi(w, w_0) \leq \epsilon$. Iterates stay in $\mathcal{B}$.

Part 2. Convergence: Telescope Lemma 6 over $i = 1, \ldots, T$. Boundary terms cancel:

$$D_\psi(w, w_0) = D_\psi(w, w_T) + \sum_{i=1}^{T} [\text{non-negative terms}]$$

Since $D_\psi(w, w_T) \geq 0$ and LHS is fixed, the sum is bounded: $T \to \infty$ forces each term to 0. In particular, $D_{\psi-\eta L_i}(w_i, w_{i-1}) \to 0$, so $w_i \to w_\infty$ (convergence).

Part 3. Interpolation: Loss terms $\eta L_i(w_i) \to 0$, so $L_i(w_\infty) = 0$ for all i. Thus $w_\infty \in \mathcal{W}$.

Proof of Theorem 4: Closeness to Optimal Point

Setup: Apply Lemma 6 to both w^* and w_∞, both in $\mathcal{W}$. Telescope over all iterations:

$$D_\psi(w^*, w_0) = D_\psi(w^*, w_\infty) + \sum_{i=1}^{\infty}[\cdots]$$

$$D_\psi(w_\infty, w_0) = D_\psi(w_\infty, w_\infty) + \sum_{i=1}^{\infty}[\cdots] = \sum_{i=1}^{\infty}[\cdots]$$

Key observation: The sums depend on w^* and w_∞ only through their distance from intermediate iterates w_{i-1}. By Assumption 2 (bounded Hessian), these distances are $O(\epsilon)$.

Conclusion: Subtracting the two equations and using Hessian bounds via Taylor expansion:

$$D_\psi(w_\infty, w_0) - D_\psi(w^*, w_0) = -D_\psi(w^*, w_\infty) + o(\epsilon)$$

Since both $D_\psi(w_\infty, w_0) - D_\psi(w^*, w_0) \geq 0$ (by definition of w^*) and $D_\psi(w^*, w_\infty) \geq 0$, each is $o(\epsilon)$.

Highly Overparameterized Regime: Why w_0 Is Close to $\mathcal{W}$

In the limit $p \gg n$, a random initialization is whp close to the interpolating set.

Linear Approximation Argument

- Distance to $\mathcal{W}$ is approximated by the solution to the linearized problem:

$$\min_{w} \|w - w_0\|^2 \quad \text{s.t.} \quad y \approx f(x, w_0) + \nabla f(x, w_0)^T (w - w_0) \tag{9}$$

- The Jacobian $\nabla f(x, w_0)$ is $n \times p$, consisting of sums of outer products
- When $p \gg n$ and data are sufficiently random: $\lambda_{\min}(\nabla f^T \nabla f) = \Omega(p)$
- Thus: $\|w^* - w_0\|^2 \approx \|y - f(x, w_0)\|^2 \cdot O(1/p) = O(n/p)$
- Conclusion: w_0 is $O(\sqrt{n/p})$ away from the manifold—negligible as p grows

Experimental Validation

Experimental Setup

Datasets: MNIST (60K samples), CIFAR-10 (50K samples). Both highly overparameterized.

Architectures: MNIST: 4-layer CNN with 2M parameters. CIFAR-10: ResNet-18 with 11M parameters.

Protocol: Random initialization $\mathcal{N}(0, 0.0001)$, train to 100% accuracy.

Seeds: 6 initializations (MNIST), 8 initializations (CIFAR-10).

Algorithms: Mirror potentials $\psi(w) = \frac{1}{q}\|w\|_q^q$ for $q \in \{1+\epsilon, 2, 3, 10\}$.

Experimental Hypothesis

The paper predicts that **each SMD algorithm finds the interpolant closest to its initialization in the corresponding Bregman divergence**.

Testing Strategy

For each initialization w_0 and each mirror type (with potential ψ):

1. Train the model to zero training error, reaching some $w_\infty \in \mathcal{W}$
2. Compute Bregman divergences $D_{\psi_q}(w_\infty^{(q)}, w_0)$ for all mirrors $q \in \{1+\epsilon, 2, 3, 10\}$
3. Check: Is the minimum always on the diagonal?

- **Prediction:** For fixed w_0 and fixed ψ_q, the closest final point to w_0 (in D_{ψ_q} distance) should be the one obtained by running SMD with ψ_q
- This would confirm both convergence to w^* and the theoretical characterization

MNIST Results: Convergence to Nearest Interpolant

../Demo/figure_mnist_distances.png

CIFAR-10 Results: Consistency at Scale

../Demo/figure_cifar_distances.png

Mirror Effects on Final Weight Distributions

../Demo/figure_weight_distributions.png

Generalization Performance Across Mirrors

MNIST: All mirrors converge to $\geq 99\%$ test accuracy. Implicit bias does not hurt MNIST generalization.

CIFAR-10: Mirror choice significantly impacts test error:

- ℓ_{10}-SMD: Best generalization (densest weights, fewest constraints)
- ℓ_2 (SGD): Intermediate performance
- ℓ_3-SMD: Slightly worse than ℓ_2
- ℓ_1-SMD: Worst (extreme sparsity overly regularizes)

Insight: Implicit bias controls generalization. Denser regularization regimes generalize better on complex tasks. Optimizer is a design lever for test performance.

Summary of Contributions

Summary

- **Implicit Bias**: Exact characterization in linear case, approximate in nonlinear
- **Star-Convexity**: Local regularity connects linear theory to deep learning
- **Optimizer as Regularizer**: Choice of ψ determines bias and generalization
- **Unified View**: SGD, exponentiated gradient, p-norm algorithms all fit framework

Generalization is the geometric footprint of the optimizer.

Future Work

1. Adaptive optimizers: What is effective ψ for Adam or RMSProp?
2. Transformer architectures: Does analysis extend to non-Euclidean geometry?
3. Tighter bounds: Can $o(\epsilon)$ be exact for specific architectures?
4. Learning ψ: Adapt mirror map during training?

Question and Answers